Liquid-side xenon transfer from a falling molten-salt droplet

Problem statement, analytical background (Newman), implementation plan, verification plan and residence-time analysis for `xenonTransferInterFoam` / `gasDiffusionSimple` (tom-v2606). All numbers in this document are computed by `src/xenonTransfer/doc/makeNewmanReport.py`.

Summary. In the falling-droplet case the gas-side resistance and the equilibrium back-pressure are negligible (ratios of order 10^{-10}), so the problem reduces to diffusion of xenon out of a sphere whose surface concentration is zero: Newman's problem [1, 2]. The thesis closes the liquid side with a constant $\mathrm{Sh}_\ell = 2$, which is the external steady value for a sphere in a stagnant medium and has no meaning for transport inside a drop. The exact inner solution removes 10.5 % of the xenon in the 11.5 ms simulated in the thesis, against 0.22 % obtained with $\mathrm{Sh}_\ell = 2$: about 48× more. The concentration boundary layer inside the drop is a few micrometres thick at these times and cannot be resolved by the VOF mesh (16–30 µm cells), so a subgrid closure $\mathrm{k}_\ell(t)$ is required. This document derives it, proposes how to carry the contact time through droplet series, re-entry and splashing, and lists what must be verified before using it.

1. Problem

1.1 Current model

`gasDiffusionSimple` computes a volumetric xenon sink in the interface cells of each identified drop:

$$\dot{m}''' = C\,K\,a_i\,\rho_\ell\,(Y_\ell - Y_{eq}), \qquad K = \left(\frac{1}{k_\ell} + \frac{k_H}{k_g}\right)^{-1}, \qquad k_\ell = \frac{Sh_\ell\,D_\ell}{d_p}$$

$$Y_{eq} = k_H\,\frac{\rho_g}{\rho_\ell}\,Y_g$$

with $\mathrm{a_i}$ the interface area density from the α = 0.5 iso-surface, Y_ℓ the liquid xenon mass fraction sampled next to the interface, and $\mathrm{Y_g}$ the gas-side value across the interface. The liquid side is closed with a `fixed` Sh_ℓ (2 in the thesis reference case, 2–100 in the sweep) or with the Higbie form used for bubbles.

1.2 Gas side and back-pressure are negligible

quantity	value
Liquid side $1/\mathrm{k}_\ell$ ($\mathrm{Sh} = 6.58$)	2.27×10^{4} s/m
Gas side $\mathrm{k_H}/\mathrm{k_g}$ ($\mathrm{Sh_g} = 2$, physical $\mathrm{D_g}$)	4.21×10^{-6} s/m
ratio gas/liquid	1.85×10^{-10}
ratio with the thesis $\mathrm{D_g} = 10^{-5}$	1.37×10^{-7}
Equilibrium back-pressure $\mathrm{Y_{eq}}/\mathrm{Y_g} = \mathrm{k_H}\rho_g/\rho_\ell$	1.32×10^{-10}

Table 1. Resistances in series for d = 0.3 mm (series resistances $1/k_\ell$ and k_H/k_g of K) and equilibrium back-pressure. Inputs:

parameter	value	source
Drop diameter d	0.30 mm	thesis
Xe diffusivity in salt D_ℓ	2.01×10^{-9} m²/s	thesis value, source to be checked
Salt density / viscosity	4125 kg/m³ / 6.07 mPa·s	thesis
He density at 920 K, 5 kPa	0.00262 kg/m³	ideal gas
He viscosity	4.69×10^{-5} Pa·s	thesis
He–Xe diffusivity D_g	0.0075 (Fuller) / 0.0073 (Chapman–Enskog) m²/s	[13, 14, 15]
Dimensionless Henry coefficient k_H	2.08×10^{-4}	thesis; cf. [16, 17]
He mean free path / Knudsen number 2λ/d	16.3 µm / 0.108	kinetic theory

The gas side therefore drops out of K for any plausible Sh_g, and the liquid sees a surface concentration of essentially zero. Note that the thesis used $D_g = 10^{-5}$ m²/s; the physical He–Xe value at 920 K and 5 kPa is about 0.0074 m²/s (0.0075 from Fuller–Schettler–Giddings [13], 0.0073 from Chapman–Enskog with Lennard-Jones parameters and the Neufeld collision integral [14, 15]). This changes nothing in K, but the same value enters the resolved gas-phase species diffusion, which is explicit in the code: a physical D_g would require $\Delta t < \Delta x^2/(6D_g) \approx 10^{-8}$ s and is not protected by `maxDi` (multiphaseInter returns a zero diffusion number).

1.3 What $Sh_\ell = 2$ means, and why it is wrong here

Sh = 2 is the steady solution for diffusion from a sphere into an unbounded stagnant medium (the outside problem). For transport inside a sphere the steady value does not exist (the drop is depleted), and the long-time limit of the bulk-referred coefficient is $2\pi^2/3 = 6.58$ [1, 2, 4]; at short times the coefficient is much larger because the depleted layer is thin (penetration regime, Higbie [3]). The values 6.58 and 17.9 in the thesis sweep are indeed the stagnant (Newman) and the circulating (Kronig–Brink [5]) long-time limits, but the transient was not considered.

1.4 Why the CFD cannot resolve it

The depleted layer grows as

$$\delta(t) \approx \sqrt{\pi D_\ell t}$$

which is 2.5 µm after 1 ms, 8.5 µm after 11.5 ms and 25 µm after 0.1 s. The droplet mesh has 16–30 µm cells; resolving the layer with a few cells needs sub-micrometre spacing, i.e. hundreds of cells per diameter in 3D. In addition the xenon storage Z = αY is advected with first-order upwind, whose numerical diffusivity ($\sim U\Delta x/2 \approx 10^{-7}$–$10^{-6}$ m²/s while the drop moves) exceeds D_ℓ by two to three orders of magnitude, so the drop interior is numerically well mixed. The thesis D_ℓ sweep confirms it: the fitted removal rate scales exactly linearly with D_ℓ (0.191, 1.91, 18.9, 183 s^{-1}), i.e. D_ℓ only acts through the input k_ℓ. Unresolved

interfacial boundary layers in VOF are a known problem and are normally handled with subgrid-scale closures [10, 11, 12]. The consistent formulation is therefore: **well-mixed drop + subgrid k_ℓ(t) driven by the drop-bulk concentration**.

2. Analytical background

2.1 Diffusion out of a sphere (Newman 1931)

For constant D_ℓ, no internal motion, uniform initial concentration c_0 and constant surface concentration c_s (here ≈ 0):

$$\frac{\partial c}{\partial t} = \frac{D_\ell}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial c}{\partial r}\right), \quad c(r,0) = c_0, \quad c(R,t) = c_s \approx 0, \quad \frac{\partial c}{\partial r}\Big|_0 = 0$$

the fraction released is [1; 2, eq. 6.20]

$$F(t) = 1 - \frac{\bar{c}(t)}{c_0} = 1 - \frac{6}{\pi^2}\sum_{n=1}^{\infty}\frac{1}{n^2}\exp\left(-n^2\pi^2 Fo\right), \qquad Fo = \frac{D_\ell t}{R^2}$$

with the short-time form [2, eq. 6.21; for Fo < 0.05 it differs from the full series by less than 1.2×10^{-10} in F and 7.0×10^{-9} in the Sh below]

$$F \simeq 6\sqrt{\frac{Fo}{\pi}} - 3\,Fo \qquad (Fo \ll 1)$$

2.2 Equivalent bulk-referred coefficient

Writing the drop balance with a coefficient referred to the bulk (volume-averaged) concentration, which is what a well-mixed CFD drop carries, gives an exact time-dependent k_ℓ:

$$\frac{d\bar{c}}{dt} = -\frac{3}{R}\,k_\ell(t)\,(\bar{c} - c_s) \quad \Rightarrow \quad k_\ell(t) = \frac{R}{3}\,\frac{dF/dt}{1-F}$$

$$Sh_\ell(Fo) = \frac{k_\ell d}{D_\ell} = \frac{2\pi^2}{3}\,\frac{\sum_n e^{-n^2\pi^2 Fo}}{\sum_n e^{-n^2\pi^2 Fo}/n^2} \;\to\; \frac{2\pi^2}{3} = 6.58 \quad (Fo \to \infty)$$

At short times this tends to the penetration result

$$Sh_\ell \simeq \frac{2}{\sqrt{\pi Fo}} \quad \Leftrightarrow \quad k_\ell = \sqrt{\frac{D_\ell}{\pi t}} \qquad \text{(Higbie penetration, } Fo \ll 1)$$

Integrating the drop balance with this k_ℓ(t) reproduces F(t) exactly, which is the key property for verification. The penetration form is within 2 % of Newman for Fo ≤ 10^{-4} and within 6 % for Fo ≤ 10^{-3} (11.5 ms for d = 0.3 mm), but 19 % low at Fo = 0.01 and a factor two low at Fo = 0.1, where depletion of the sphere interior matters (table 2). This matters for splashing and films (section 3.2), where the sphere geometry is lost but the surface-age concept survives only in the penetration regime.

Fo	Sh_ℓ Newman	$2/(\pi Fo)^{1/2}$	ratio
1.00×10^{-4}	114.7	112.8	1.016
0.001	37.59	35.68	1.054
0.01	13.43	11.28	1.190
0.03	8.963	6.515	1.376
0.1	6.834	3.568	1.915

Table 2. Newman bulk-referred Sh vs. the penetration limit.

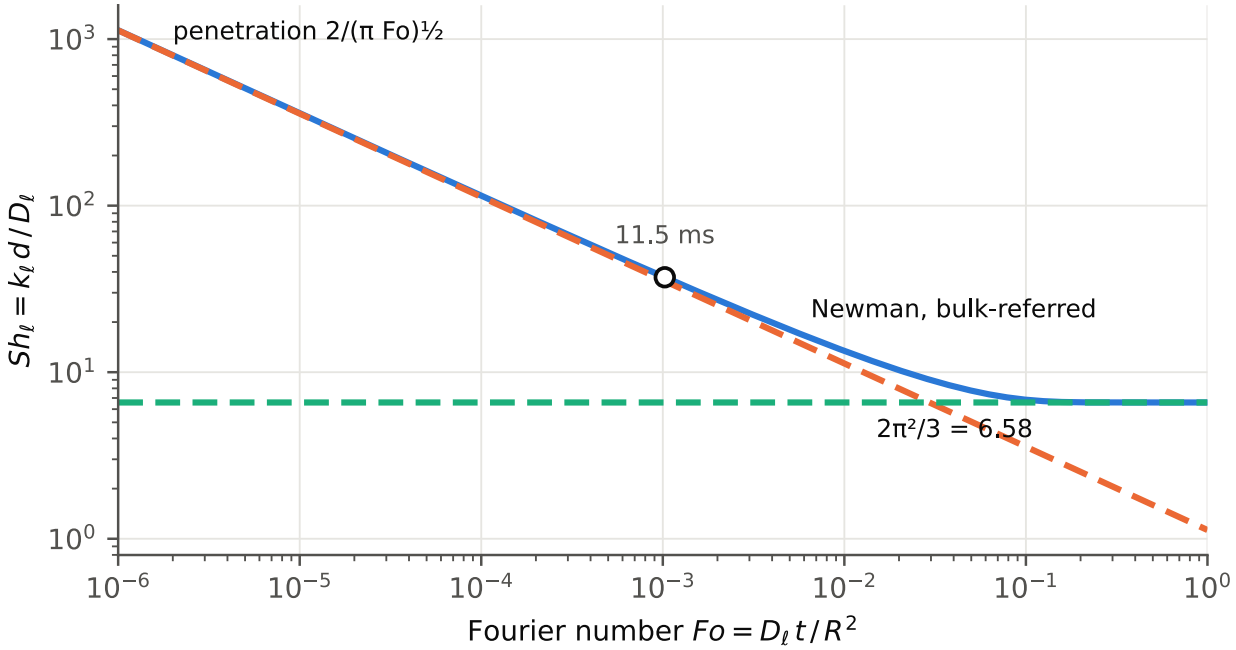


Figure 1. Bulk-referred liquid-side Sherwood number of a stagnant sphere. Marker: the end of the thesis simulation (Fo = 0.00103, Sh = 37.2).

2.3 Consequences for the thesis case

t	Fo	Newman F	Sh = 2 (thesis)	Sh = 6.58	Sh_ℓ(Fo)	$\delta = (\pi D_\ell t)^{1/2}$
1 ms	8.92×10^{-5}	3.17 %	0.0267 %	0.088 %	121	2.51 µm
5 ms	4.46×10^{-4}	7.01 %	0.134 %	0.439 %	55.3	5.61 µm
11.5 ms	0.00103	10.5 %	0.307 %	1.01 %	37.2	8.51 µm
32.27 ms	0.00288	17.3 %	0.859 %	2.8 %	23	14.3 µm
100 ms	0.00892	29.3 %	2.64 %	8.42 %	14.1	25.1 µm
1000 ms	0.0892	74.3 %	23.5 %	58.5 %	6.93	79.4 µm

Table 3. Removed fraction for d = 0.3 mm. “Sh = 2” and “Sh = 6.58”: well-mixed drop with constant coefficient, $F = 1 - \exp(-6\ Sh\ D_\ell\ t/d^2)$. The thesis CFD obtained 0.22 % at 11.5 ms with Sh = 2 (slightly below 0.31 % because of the area/diameter reconstruction issue of section 5).

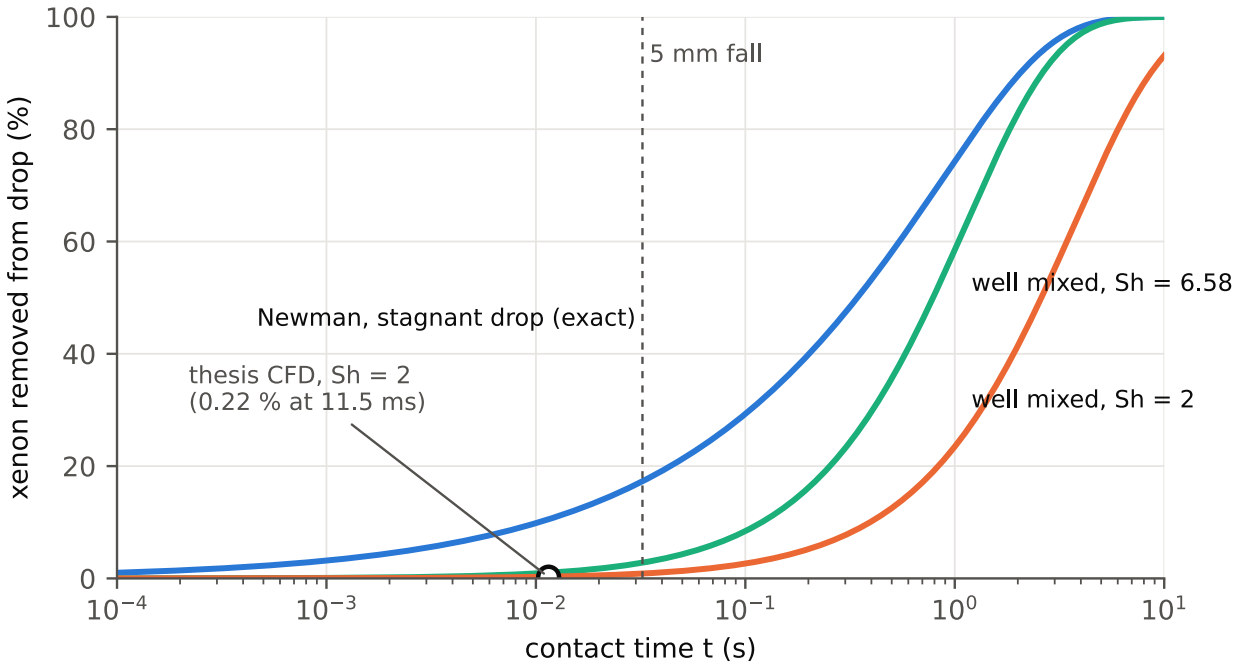


Figure 2. Xenon removed from a 0.3 mm drop vs. contact time: exact stagnant-drop solution and well-mixed models with constant Sh.

2.4 Validity conditions (to be checked, not assumed)

- **No internal circulation.** For a drop in a gas the viscosity ratio is $\kappa = \mu_\ell/\mu_g = 129$, and the Hadamard–Rybczynski interface velocity of a clean drop is U/(2(1+κ)) [6, 7]. The internal Péclet number is

$$Sh_\infty^{KB} \approx 17.9, \qquad Pe_{int} = \frac{U\,d}{2(1+\kappa)\,D_\ell}, \quad \kappa = \mu_\ell/\mu_g$$

  and the number of internal turnovers after a fall z is $N \approx z/((1+\kappa)\pi d)$. At the end of the thesis window (U = 0.11 m/s) $Pe_{int} = 64$ but N = 0.0052; after 5 mm N = 0.041; after the 0.46 m needed for 50 % removal N = 3.8; after 7.5 m (90 %) N = 61. Circulation is therefore irrelevant for millimetre falls, but for decimetre-to-metre contactors it can develop and push transfer towards the Kronig–Brink limit ($Sh_\infty \approx 17.9$, effective diffusivity 2.25 D_ℓ [5, 7]). Newman is the lower bound and Kronig–Brink the laminar upper bound; surface-active impurities push towards Newman. The solver already prints the resolved internal-circulation velocity and Péclet number (`debugInternalCirculation`), but see the parasitic-current caveat in section 5.
- **Uniform initial concentration** and **constant D_ℓ** (dilute xenon, isothermal drop).
- **Near-spherical drop.** For deformed drops the area enters through A_j; in the penetration regime the flux per unit area does not depend on the shape.
- **Zero surface concentration**, verified in section 1.2.
- **Rarefaction.** The helium mean free path is 16 µm (Knudsen number $2\lambda/d = 0.11$); this only affects the drag (Cunningham factor 1.136), not the liquid side.

3. Implementation plan (not yet implemented)

3.1 Closure

New option `liquidSherwoodModel Newman;` in `gasDiffusionSimple`. For each drop j:

$$\dot{M}_j = k_\ell(Fo_j)\, A_j\, \rho_\ell \left(\bar{Y}_{\ell,j} - \bar{Y}_{eq,j}\right), \qquad Fo_j = \frac{4 D_\ell\, \theta_j}{d_j^2}$$

where θ_j is the contact time of drop j, A_j its transfer area (already available: `dropArea iso | volumeSphere` with `correctDropArea`), and $\bar{Y}_{\ell,j}$ the mass-weighted bulk xenon fraction of the drop (already computed as `dropYbulk`). The drop flux is distributed over the drop's flux cells proportionally to their area, so the existing conservative face-based mapping to the gas is reused unchanged. Sh(Fo) is evaluated with the short-time expression for Fo < 0.05 and the series (truncated where $n^2\pi^2 Fo > 50$) above.

Driving the flux with the bulk value instead of the interface-cell value removes the mesh dependence of the sampled concentration and makes the CFD drop reproduce the Newman curve exactly in the limit where it applies.

3.2 Contact time for single drops, series, re-entry and splashing

The target applications (a series of droplets, the same droplet re-entering the domain after leaving it, splashing on an inclined plate) rule out a global clock and make Lagrangian drop-identity tracking fragile (drops merge, split, leave and re-enter, and identities are not preserved by the region-split clustering between steps or across processors). The proposal is an Eulerian **contact-age** field θ carried by the liquid, transported with the same conservative Z = αθ machinery as xenon (mass-consistent with α):

$$\frac{\partial(\alpha_\ell\, \theta)}{\partial t} + \nabla \cdot (\alpha_\ell\, \phi\, \theta) = \alpha_\ell\, \chi_{drop}$$

with χ_{drop} = 1 in cells belonging to an identified drop (or film fragment) in contact with helium and 0 in the feed/injector liquid. This is the "local mean age" concept of residence-time distributions [8, 9]. The drop value is the mass-weighted mean θ_j. Behaviour in the target cases:

- **Series of drops:** each drop starts with θ = 0 at the injector and accumulates its own age; no identity bookkeeping.
- **Re-entry:** if the outlet is mapped back to the inlet, both Z_{Xe} and Z_θ are carried, so the drop resumes with its own depletion history (F continuous across the boundary).
- **Breakup:** fragments inherit the parent age (exact to leading order in the penetration regime, where the depleted layer stays on the surface).
- **Coalescence:** mass-weighted mean age; this is an approximation because $k_\ell(\theta)$ is non-linear (state it as such).
- **Splashing / film formation:** new interface exposes undepleted liquid (surface renewal, Danckwerts [4]). Proposed per-drop renewal when the drop area grows:

$$\theta_j^{n+1} = \left(\theta_j^n + \Delta t\right) \min\left(1, \frac{A_j^n}{A_j^{n+1}}\right)$$

i.e. the newly created area is assigned zero age and mixed by area with the old surface. On the plate the sphere geometry is lost; the penetration form $k_\ell = (D_\ell/\pi\theta)^{1/2}$ remains valid while the depleted depth $\delta = (\pi D_\ell \theta)^{1/2}$ is small compared with the film thickness; beyond that a film model is needed. This renewal rule is a modelling hypothesis and must be validated, not only verified.

3.3 Numerical aspects

- The source stays explicit: $\beta\Delta t = 6\mathrm{Sh}\, D_\ell \Delta t/d^2 < 10^{-3}$ even at $\mathrm{Fo} = 10^{-6}$.
- Per-drop quantities need global reductions; for droplet series the ~20 scalar reductions per drop per step should be merged into a single list reduction.
- Sh(Fo) is singular as $\mathrm{Fo} \to 0$; the first step after release should use the time-averaged coefficient over the step, $\bar{k} = (F(\theta+\Delta t) - F(\theta))/(\Delta t \cdot 3(1-F(\theta))/R)$, which is exact for the step.

4. Verification

4.1 Why

The closure replaces physics the mesh cannot resolve. The CFD then only supplies geometry (area, volume), motion and bookkeeping (mass conservation, mapping to the gas, age transport). Verification checks that this chain reproduces the analytical solution exactly where that solution applies, so that later differences in the target cases (series, re-entry, splashing) can be attributed to physics rather than to implementation or discretisation. It is distinct from validation against experiments, which concerns whether Newman/renewal is the right physics (e.g. internal circulation, section 2.4).

4.2 What to verify

#	test	checks	acceptance
V0	Unit test of Sh(Fo), F(Fo)	branch switch, limits $2/(\pi\mathrm{Fo})^{1/2}$ and 6.58, against an independent series	relative error $< 10^{-6}$
V1	0D integration of the drop balance with the solver time stepping	explicit source and first-step treatment reproduce F(t)	error < 0.1 % in F up to Fo = 0.1
V2	Stagnant drop in quiescent helium (g = 0), three meshes	area, bulk sampling, mapping to the gas, conservation, mesh independence	F within 2 % of Newman at 10, 50, 100 ms on all meshes; liquid loss = $\int \dot{m}$ within 0.1 %
V3	Falling drop (this case) to t ≈ 35 ms (5 mm)	insensitivity to motion and slip; internal-circulation diagnostics	same F as V2 within 2 %; $\mathrm{Pe}_{\mathrm{int}}$ reported
V4	Two drops released at different times	age transport and per-drop θ_j	θ_j equals each drop's own contact time within one step
V5	Drop leaving the bottom and re-entering at the top (mapped boundary)	conservation of Z_{Xe}, Z_θ across the mapping	F(t) continuous, total Xe conserved to 10^{-8}
V6	Prescribed stretching of a drop (kinematic test)	area-renewal rule	θ_j follows the renewal rule exactly

Table 6. Verification tests, in the order they should be run.

V2 is the central test: with g = 0 the Newman solution is exact for the CFD set-up, so any discrepancy is an implementation or geometry error. Before the Newman model, the same test with a fixed Sh already checks the geometry chain (section 5).

5. Geometry and slip fixes already applied (for reference)

Measured on `run/xenonDroplet` with the original code (Sh = 2):

quantity	t = 0 (sharp interface)	t ≥ 40 µs (smeared)
Iso-surface area / sphere area	0.71	1.14
d_p = 6V/A (true 0.30 mm)	0.418 mm	0.26 mm
Fraction of iso area that carries flux	1.00	0.61
Drop removal rate β (nominal $6ShD/d^2$ = 0.267 s^{-1})	0.138 s^{-1}	0.213 s^{-1}

The flux scales as $k_\ell A \propto A^2/V$, so area errors enter squared. The face-based mapping to the gas is conservative (liquid loss / integrated model rate = 0.9995, total xenon conserved to 2.50×10^{-10}). Fixes now in the code: `dpMode perDropVol` (diameter from the conserved volume), `dropArea volumeSphere|iso` with `correctDropArea` (each drop transfers through its full area), `slipVelocity shell` (drop velocity minus helium velocity averaged in a shell 1.5–3 radii away; the phases share one velocity field, so the old adjacent-cell difference was not a slip velocity), and Y = Z/α reconstructed with the start-of-step α.

With the fixes (`dpMode perDropVol`, `dropArea volumeSphere`, `correctDropArea true`, `slipVelocity shell`) the same run gives a drop removal rate, measured from the liquid xenon inventory, of β = 0.2709 s^{-1} against the nominal 0.272 s^{-1} ($6\cdot Sh\cdot D_\ell/D_{vol}^2$), constant to 0.3 % while the interface smears; the flux cells are rescaled by 1.40–1.60. Liquid loss / integrated model rate = 0.99996.

> **Parasitic currents.** At 0.3 ms the maximum velocity in the domain is 0.36 m/s, in the helium just above the drop, while the drop falls at 0.00156 m/s. The capillary velocity σ/μ_g is about 5578 m/s and the density ratio about 1.6×10^6, so spurious currents in the gas are expected. As a consequence the shell slip velocity is 0.0079 m/s (the old adjacent-cell value was 0.00165 m/s): the shell definition is correct, but its input is polluted at these low drop velocities. This does not affect the present results (the slip only enters the negligible gas side) but it must be reduced before any velocity-dependent closure (internal circulation, Higbie, Kronig–Brink switching) is trusted: a static-drop test (g = 0) quantifies it directly.

6. Residence time and fall height

6.1 Drop kinematics

The drop is released from rest. With Schiller–Naumann drag and the Cunningham slip correction C_c:

$$\frac{dv}{dt} = g\left(1 - \frac{\rho_g}{\rho_\ell}\right) - \frac{v\,(1 + 0.15\,Re^{0.687})}{\tau_p}, \qquad \tau_p = \frac{\rho_\ell d^2}{18\,\mu_g}\,C_c$$

d	τ_p	terminal velocity	Re at terminal velocity	t to fall 5 mm
0.1 mm	0.069 s	0.675 m/s	0.00376	34.6 ms
0.3 mm	0.5 s	4.77 m/s	0.0799	32.3 ms
1 mm	5.08 s	39.5 m/s	2.2	32 ms

Table 4. Fall kinematics in helium at 920 K, 5 kPa.

Because τ_p = 0.5 s for d = 0.3 mm, the drop is in practically free fall over the thesis window: it falls only 0.64 mm in 11.5 ms, so the 5 mm column is never traversed; reaching 5 mm takes 32.3 ms. In the thesis set-up the drop also starts 0.35 mm (about 12 cells) below the top boundary. Since transfer is controlled by the liquid side and there is no internal circulation in this window, what matters is the contact time, not the height. Note that the drop never reaches its terminal velocity in any practical column (τ_p = 0.5 s, terminal velocity 4.8 m/s at Re = 0.08), so the free-fall height grows almost as ½gt² up to a few tenths of a second.

6.2 Required contact time and height

d	target F	Fo	contact time	free-fall height	thesis extrapolation (Sh = 2, ½gt²)
0.1 mm	50 %	0.0305	0.0381 s	0.00596 m	
0.1 mm	90 %	0.183	0.228 s	0.109 m	
0.1 mm	99 %	0.416	0.519 s	0.304 m	
0.3 mm	50 %	0.0305	0.343 s	0.463 m	3.63 s / 64.6 m
0.3 mm	90 %	0.183	2.05 s	7.53 m	12.1 s / 713 m
0.3 mm	99 %	0.416	4.67 s	20 m	24.1 s / 2.85e+03 m
1 mm	50 %	0.0305	3.81 s	54.7 m	
1 mm	90 %	0.183	22.8 s	> 20 s of fall	
1 mm	99 %	0.416	51.9 s	> 20 s of fall	

Table 5. Contact time from Newman and the corresponding free-fall height (drag and slip included). Last column: the thesis back-up estimate, i.e. first-order decay with the fitted β = 0.191 s^{-1} (Sh = 2) and drag-free fall H = ½gt².

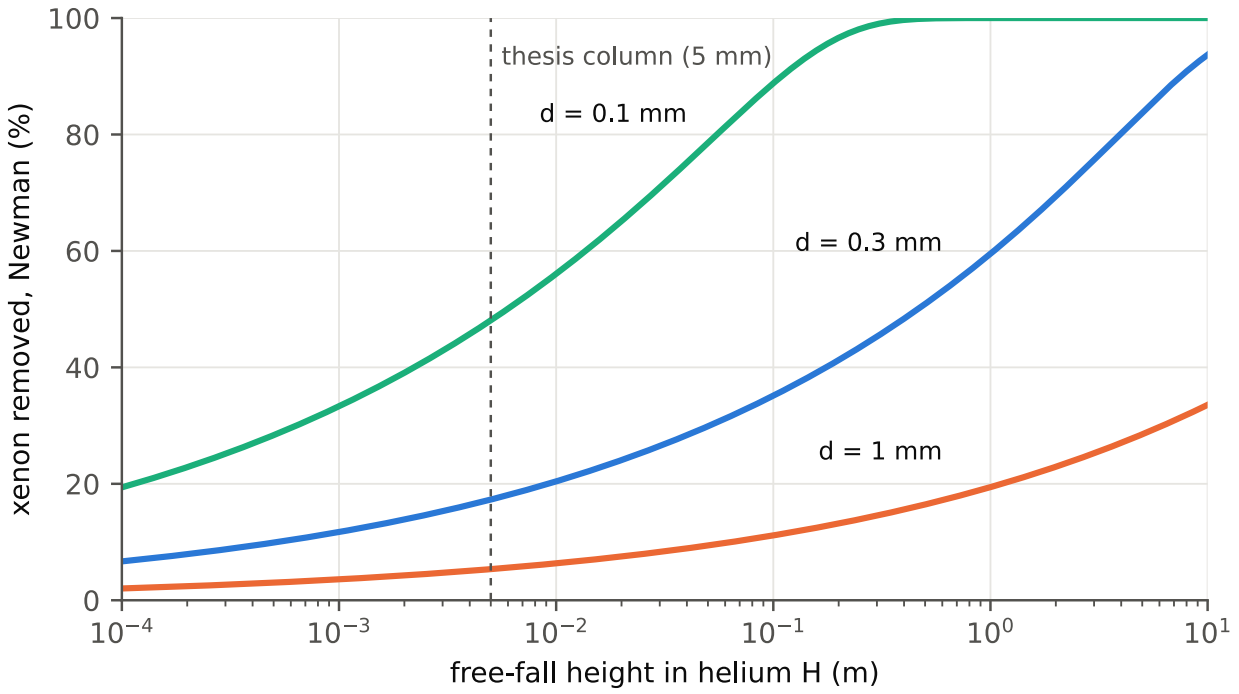


Figure 3. Xenon removed vs. free-fall height in helium for three drop diameters (Newman, stagnant drop). Small drops strip faster (Fo ∝ t/d²) but also fall slower.

> The thesis extrapolation underestimates removal at short contact times by more than an order of magnitude (wrong Sh and an exponential instead of a √t law) and assumes drag-free fall at all heights. Conversely, at long contact times possible internal circulation would increase transfer above Newman. Contactor heights should be estimated with the Newman curve as a conservative reference and bracketed with Kronig–Brink once the circulation diagnostics are available.

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Bibliographic details were written from memory and should be checked against the originals before citing.

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